

Common Reward Function Pitfalls

Reward
Hacking

Mode
Collapse

Metric
Gaming

Validation
Overfitting

Detect: $\Delta_{\text{overfit}} = \text{mAP}_{\text{val}} - \text{mAP}_{\text{test}}$

Detect: Coverage(t) declining over time

Detect: $\bar{N}_{\text{pred}}$ abnormally high

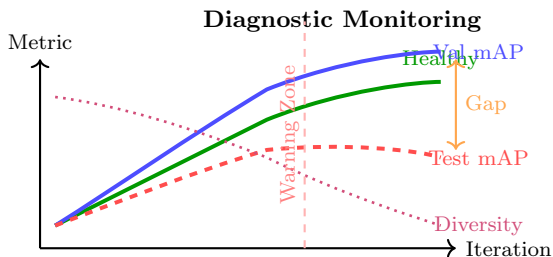
Detect: Val-holdout gap increasing

Mitigate: Holdout monitoring, Regularization

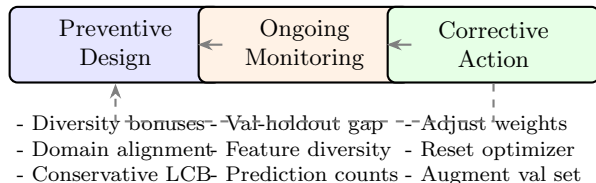
Mitigate: Diversity bonus, DPP reward

Mitigate: Prediction count penalty

Mitigate: Early stopping, Conservative LCB



Mitigation Framework



Iterative Refinement